

GRAY AFF, WA, USA

WMO: 742071

Lat: 47.083N

Lon: 122.583W

Elev: 91

StdP: 100.23

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24201

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-6.2	-4.0	-10.2	1.6	-2.4	-7.2	2.1	-2.0	10.4	9.7	9.0	8.9	1.5	140	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	31.0	19.0	28.7	18.2	26.9	17.5	20.0	28.4	19.0	26.7	18.2	25.1	3.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	12.5	21.1	16.3	11.7	20.4	15.5	11.1	20.0	57.1	28.4	54.0	26.8	51.3	25.1	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	6.0	-9.1	34.4	2.7	2.0	-11.0	35.9	-12.6	37.1	-14.1	38.3	-16.0	39.7		
(5)			-9.6	21.3	2.9	1.0	-11.7	22.1	-13.4	22.7	-15.1	23.3	-17.2	24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	4.6	5.2	7.2	9.3	12.8	15.6	18.7	18.7	15.7	10.7	6.8	3.8	(6)
(7)		DBStd	6.02	3.40	3.08	2.74	2.74	3.06	2.86	2.66	2.37	2.64	2.76	3.52	3.80	(7)
(8)		HDD10.0	778	169	137	96	44	8	0	0	0	1	24	107	194	(8)
(9)		HDD18.3	2843	426	368	346	270	172	93	27	24	84	236	347	450	(9)
(10)		CDD10.0	1066	2	1	8	24	96	167	268	269	173	46	10	2	(10)
(11)		CDD18.3	91	0	0	0	0	2	10	37	36	7	0	0	0	(11)
(12)		CDH23.3	1442	0	0	1	11	69	184	540	482	154	2	0	0	(12)
(13)		CDH26.7	451	0	0	0	2	13	62	191	151	33	0	0	0	(13)
(14)	Wind	WSAvg	2.6	2.5	2.6	2.9	2.9	2.7	2.8	2.6	2.5	2.4	2.4	2.6	2.5	(14)
(15)	Precipitation	PrecAvg	1149	166	117	125	85	56	42	16	27	44	112	180	167	(15)
(16)		PrecMax	1441	346	280	242	202	100	92	42	95	211	276	410	321	(16)
(17)		PrecMin	850	72	3	35	27	3	4	2	1	1	15	44	43	(17)
(18)		PrecStd	186	62	59	49	45	28	23	12	24	45	56	87	72	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	15.1	20.1	24.8	28.7	32.3	34.1	32.9	30.1	22.5	17.5	14.6	(19)
(20)			MCWB	12.3	10.7	12.5	14.9	16.6	19.1	20.0	19.6	18.2	14.7	14.4	12.4	(20)
(21)		2%	DB	12.3	13.0	16.5	20.3	25.2	28.0	31.2	30.2	27.3	19.7	14.9	12.2	(21)
(22)			MCWB	10.7	10.0	10.8	12.6	15.6	17.7	19.1	19.0	17.2	14.3	13.0	10.9	(22)
(23)		5%	DB	10.9	11.7	14.1	17.6	22.5	25.0	28.7	28.0	25.0	17.5	13.1	10.6	(23)
(24)			MCWB	9.6	9.1	10.0	11.3	14.5	16.3	18.5	18.1	16.7	13.4	11.7	9.6	(24)
(25)		10%	DB	9.7	10.2	12.5	15.1	19.9	22.4	26.3	26.1	22.7	16.0	12.2	9.0	(25)
(26)			MCWB	8.7	8.3	9.3	10.1	13.7	15.3	17.4	17.5	15.8	13.0	11.1	8.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	12.8	13.7	15.8	18.3	20.4	21.5	21.1	20.1	17.0	15.4	12.9	(27)
(28)			MCD B	14.3	13.8	17.5	23.0	24.7	29.5	31.8	29.5	25.7	19.0	16.4	14.4	(28)
(29)		2%	WB	11.2	11.1	12.0	13.6	16.6	18.5	20.0	19.8	18.5	15.5	13.4	11.1	(29)
(30)			MCD B	11.7	11.9	14.8	18.3	22.7	26.4	29.2	28.0	24.6	17.7	14.5	11.9	(30)
(31)		5%	WB	9.8	9.6	10.9	12.2	15.3	16.9	18.9	18.9	17.5	14.4	12.3	9.6	(31)
(32)			MCD B	10.4	10.7	13.2	16.0	21.7	23.4	27.2	26.5	22.8	16.5	13.1	10.3	(32)
(33)		10%	WB	8.6	8.5	9.9	11.1	14.1	15.8	18.0	18.0	16.6	13.4	11.1	8.4	(33)
(34)			MCD B	9.2	9.8	12.1	14.3	19.1	21.4	25.2	24.8	21.4	15.5	11.9	8.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.8	8.4	9.5	10.5	11.2	11.3	13.0	13.1	12.5	9.7	7.9	6.4	(35)
(36)			MCD BR	7.9	10.8	12.8	15.4	17.3	17.1	18.2	17.7	17.3	12.7	8.4	6.7	(36)
(37)			MCWBR	6.0	7.3	7.8	8.3	8.6	7.8	7.7	7.7	8.6	7.9	6.4	5.4	(37)
(38)		5% WB	MCD BR	6.7	8.0	10.6	12.8	14.5	14.6	15.9	15.3	14.2	9.9	7.2	6.2	(38)
(39)			MCWBR	5.7	6.0	7.1	7.4	7.5	7.0	7.1	6.9	7.5	6.8	5.9	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.325	0.314	0.328	0.355	0.375	0.364	0.349	0.354	0.342	0.330	0.324	0.331	(40)	
(41)		taud	2.479	2.540	2.521	2.419	2.380	2.435	2.499	2.488	2.521	2.558	2.515	2.394	(41)	
(42)		Ebn at Noon	763	864	901	901	889	899	909	891	873	832	759	696	(42)	
(43)		Edh at Noon	66	75	89	108	116	111	102	99	88	73	63	64	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.03	1.88	2.79	4.09	5.18	5.67	6.33	5.42	3.99	2.26	1.21	0.85	(44)	
(45)		RadStd	0.16	0.26	0.34	0.39	0.44	0.58	0.48	0.38	0.37	0.26	0.16	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.99	+1.11	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page